

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

PRINCIPLES OF COMMUNICATION SYSTEMS

(Common to CSE, IT, AI&DS, CSE(AI), CSE(DS), CSE(AI&ML), CSE(IoT), CSE(CS), CS&D and AI&ML)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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| (a) Define SNR and figure of merit of a communication system. | 2M |
| (b) Draw the square law modulator. | 2M |
| (c) With necessary equation, define transmission efficiency of an AM wave. | 2M |
| (d) Why pre-emphasis and de-emphasis are needed in F.M but not in A.M? Explain. | 2M |
| (e) Define TDM. | 2M |
| (f) Why noise immunity of PWM is better than that of PAM? | 2M |
| (g) Draw the power spectral density of BFSK signal. | 2M |
| (h) What is signal constellation diagram and describe its purpose? | 2M |
| (i) List the applications of RADAR. | 2M |
| (j) What is frequency reuse? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 With block diagram, explain the principle of operation of radio transmitter and receiver. 10M
- OR**
- 3 (a) Define Amplitude modulation and explain about modulation index. 5M
(b) Explain the principle of V.S.B Transmission. What are its advantages over S.S.B? 5M
- 4 (a) Derive the bandwidth of tone modulated FM signal and the effect of modulation index on bandwidth. 5M
(b) Draw the phasor diagram of narrow band FM signals with equation. 5M
- OR**
- 5 Formulate the equation for FM wave. Define modulation index, maximum deviation and bandwidth of a FM signal. 10M
- 6 Explain, how a PPM signal can be generated from PWM signal. 10M
- OR**
- 7 Define Pulse Amplitude Modulation (PAM)? Give merits and demerits of PAM. 10M
- 8 Explain in detail about Quadrature phase shift keying. 10M
- OR**
- 9 (a) Discuss the generation and demodulation of binary frequency shift keying. 5M
(b) Illustrate the principle of regenerative repeater. 5M
- 10 Write about optical communication with block diagram. 10M
- OR**
- 11 (a) Explain the principle of operation of Radar system. 5M
(b) Write short notes on TDMA and CDMA. 5M
